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EFFECTIVENESS OF EXERCISE ON OSTEOSARCOPENIA IN ELDERLY PEOPLE: A PROTOCOL SYSTEMATIC REVIEW

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-045604
Article Type:	Protocol
Date Submitted by the Author:	06-Oct-2020
Complete List of Authors:	Vinícius-Souza, Guilherme; Universidade Federal de Goias, Faculdade de Medicina Silveira, Erika; Universidade Federal de Goias, Faculdade de Medicina Noll, Matias; Instituto Federal Goiano; University of Southern Denmark
Keywords:	GERIATRIC MEDICINE, Musculoskeletal disorders < ORTHOPAEDIC & TRAUMA SURGERY, SPORTS MEDICINE

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EFFECTIVENESS OF EXERCISE ON OSTEOSARCOPENIA IN ELDERLY PEOPLE: A PROTOCOL SYSTEMATIC REVIEW

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EFFECTIVENESS OF EXERCISE ON OSTEOSARCOPENIA IN ELDERLY PEOPLE: A PROTOCOL SYSTEMATIC REVIEW

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ABSTRACT

Introduction: Osteosarcopenia is defined as the concomitant occurrence of sarcopenia and osteopenia or osteoporosis. Older adults with this syndrome have greater fragility and chances of mortality compared to those without these conditions. Exercise has been recommended as a treatment for osteosarcopenia based on interventions with sarcopenic and osteoporotic individuals separately. However, there is no evidence that physical exercise can really be an effective treatment for osteosarcopenia. Our objective is to identify whether physical exercise can improve the osteosarcopenia in older adults and lead to good health outcomes. **Methods and analysis**: We will perform a systematic review on the follow databases: PubMed, Embase, Cochrane, and Scopus. The criterion of inclusion will be clinical trial studies in which the interventions were physical exercises in older adults diagnosed with osteosarcopenia. To assess the risk of bias, the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) and the Black and Downs tools will be used. For each search result, the quality of the evidence will ultimately receive one of four grades: high quality, moderate quality, low quality, or very low quality. The outcome of this study will be the improvement in osteosarcopenia (bone mineral density, appendicular muscle mass, muscle strength, and function) in elderly people. The possibility of meta-analysis will be assessed according to the homogeneity of the studies, using the methods of fixed or random effect. Sensitivity analyses will be performed, and the funnel plot will be used to assess publication bias. The proposed statistical analyses will be performed using the STATA Software, version 14.0. **Ethics and dissemination**: The results of the systematic review will be disseminated via

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publication in a peer-reviewed journal and presented at a relevant conference. The data we will use do not include individual patient data, so ethical approval is not required.

Systematic review registration: Ongoing on Prospero.

Keywords: Older adults. Osteoporosis. Sarcopenia.

Strengths and limitations of this study

- The prescription of PE has been described based on specific recommendations for osteoporosis and sarcopenia and there is no systematic review specifically with osteosarcopenia in the elderly.
- Our investigation will seek to identify and determine which duration, frequency, intensity, and type of PE is the most appropriate for preventing and treating osteossarcopenia.
- Our review will include the number of studies included as the literature is still scarce on the subject

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Introduction

Osteosarcopenia is a recent syndrome [1], defined as the presence of sarcopenia plus osteopenia or osteoporosis in the same individual [2]. The diagnosis of osteopenia / osteoporosis provides an assessment of bone mineral density (BMD) through the absorption of double energy by X-rays (DEXA) [3,4]. Sarcopenia, on the other hand, is characterized by low muscle strength, with the diagnosis confirmed through the detection of low muscle quality and it is identified as severe sarcopenia through weak physical performance [5].

Osteosarcopenia in elderly people in Germany had a prevalence of 28% [6], while in a study carried out with elderly people from the Australian community, the rates were 37% [7]. In addition, when compared to the presence of isolated conditions, i.e., sarcopenia, osteopenia, or osteoporosis, elderly people with osteosarcopenia have more fractures [8] and a 15.1% higher chance of mortality [9].

Physical exercise (PE) has been recommended in recent reviews as a prevention strategy or non-pharmacological therapeutic approach for osteosarcopenia [10,11]. However, the prescriptions prescribed for sarcopenia and osteoporosis alone [11–13], consider that there is a consensus in the literature about the effectiveness of PE for elderly people with sarcopenia and osteoporosis [14–17].

Regarding osteosarcopenia, no literature review was found that addressed the effects of regular PE practice on osteosarcopenia for elderly people. Therefore, our aim is to develop a systematic review of the literature that can answer the following question: "What are the influences of regular PE practice on osteosarcopenia in elderly people?" The results of this study may promote the application of PE as a form of non-

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pharmacological or complementary therapy for the prevention or treatment of osteosarcopenia.

Methods and analysis

This systematic review will be prepared according to the criteria of the Preferred Reporting Items for Systematic review and Meta-Analysis Protocols - PRISMA-P [18,19]. We adopted the "PICO" structure and defined it as follows: "P" represents elderly people, "I" represents physical exercise, "C" represents without physical exercise, and "O" represents osteosarcopenia.

Eligibility criteria

Controlled, randomized, blind, or open clinical trials will be included that performed an intervention with any modality of physical exercise with a minimum time of four weeks, conducted with elderly people (age ≥65 years) of both sexes, diagnosed with osteosarcopenia, i.e., individuals with osteopenia (T-score - 2.5 to <-1) and osteoporosis (T-score ≤ -2.5) according to the World Health Organization (WHO) diagnostic criteria [3], and sarcopenia according to a grant from the European Working Group on Sarcopenic in Older People (EWGSOP) [5,20], without language restrictions and period of publication.

Studies with hospitalized elderly people, which focused on specific conditions, e.g., stroke, will be excluded. Observational studies, opinion articles, editorials, narrative reviews, case series and comments, duplicate studies and publications that present data unavailability even after asking the authors.

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Database

Our search strategy will be conducted in four databases (PubMed (National Library of Medicine), Embase, Cochrane, and Scopus). In addition, the GreyNet International platform will be used to locate productions referring to gray literature internationally as well as Google Scholar. To complement the search and ensure the saturation of the literature, the references of the selected articles will also be considered.

Search terms were selected using the MeSH (Medical Subject Headings) for the PubMed database. Search: ((Osteosarcopenia [Title/Abstract]) AND (((((exercise [MeSH Terms]) OR (exercise [Title/Abstract])) OR (physical activity [Title/Abstract])) OR (physical activities [Title/Abstract])) OR ("Physical Fitness"[Title/Abstract]))) AND ((((((((aged[MeSH Terms]) OR (aged[Title/Abstract])) OR (elderly[Title/Abstract])) OR (older adult [Title/Abstract])) OR (older adults [Title/Abstract])) OR (elder[Title/Abstract])) OR (Ageing[Title/Abstract])) OR (aging[Title/Abstract])). The same search strategy model was adapted to Embase, Cochrane, and Scopus databases.

Review Process

The search for articles will be carried out by two independent researchers (GVS, MN) and a third senior reviewer (EAS). The inclusion of articles will be carried out by reading the titles, and later by reading the abstracts. Finally, the complete content will be analyzed for inclusion.

After executing the search strategy, articles will be collated, and duplicates will be removed using Mendeley Software. Then, two reviewers (GVS and MN) will independently screen the titles and abstracts of all articles identified in the literature

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search for inclusion. Disagreement regarding inclusion will be discussed and resolved by a third reviewer (EAS). The screening process will be performed for both reviewers using Rayyan Software [21]. Inter-rater reliability for individual component ratings will be determined by calculating the percentage of agreement and the Cohen's Kappa coefficient [22]. The remaining articles will be read in full and evaluated to determine their eligibility based on the inclusion and exclusion criteria. Finally, the eligible articles will be included in the systematic review.

We will prepare a flowchart with information about the screening of studies, the included studies and the reasons for excluding others, the recording and viewing of this process, following the recommendation of PRISMA-P.

Data extraction and study quality assessment

To extract data from an article, a standardized form prepared by the authors will be used. The following items will be considered: author / year of publication, place of study, age group studied, sample size, intervention performed, time of intervention, and main results.

To assess the risk of bias, the Grading of Recommendations, Assessment, Development and Evaluations - GRADE [23] and Black and Downs [24] tools will be used. For each research result, the quality of the evidence will ultimately receive one of four scores: high quality, moderate quality, low quality, or very low quality [25]. We will also analyze whether the authors of the included studies addressed the impact of possible conflicts of interest and information regarding ethical approval [26].

Data analysis

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The outcome of this study will be the improvement in osteosarcopenia (bone mineral density, appendicular muscle mass, muscle strength, and function) in elderly people. The possibility of meta-analysis will be assessed according to the homogeneity of the studies, using the methods of fixed or random effect. The Chi-square test will be applied to assess heterogeneity, with a significance level of $p < 0.05$. The I-square (I^2) statistic will be used to assess the magnitude of inconsistency, which will point to high heterogeneity when the results are greater than 75%, moderate heterogeneity when the results are between 25% to 75%, and an I^2 less than 25% will demonstrate low heterogeneity [27,28]. Sensitivity analyses will be performed, and the funnel plot will be used to assess publication bias. The proposed statistical analyses will be performed using the STATA Software, version 14.0.

Discussion

The prescription of PE has been described based on specific recommendations for osteoporosis and sarcopenia [10,11]. This is why our investigation will seek to identify and determine which duration, frequency, intensity, and type of PE is the most appropriate for preventing and treating osteosarcopenia. We will seek to broaden the research by addressing the possible evidence of different exercise modalities: low or high impact aerobics, resistance exercise, balance exercises, combined exercises, and whole body vibration.

By carrying out this protocol, a systematic review and possible meta-analysis to elucidate which physical exercises are most effective for the treatment of osteosarcopenia in elderly people will be produced, as well as whether to prevent the problem. Until then, we are not aware of a published systematic review on this topic. Consequently, the results

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may bring important recommendations for the field of Gerontology. At the conclusion of this project, we aim to clarify the influence of physical exercise on the parameters of osteosarcopenia, to trace paths of recommendations for a possible therapeutic practice, and to identify the need for new studies.

Ethics and dissemination

The results of the systematic review will be disseminated via publication in a peer-reviewed journal and presented at a relevant conference. The data we will use do not include individual patient data, so ethical approval is not required.

Abbreviations: BMD – Bone mineral density; DEXA – Absorption of double energy by X-rays; PE – Physical exercise; PRISMA-P – Preferred Reporting Items for Systematic review and Meta-Analysis Protocols; WHO – World Health Organization; EWGSOP – European Working Group on Sarcopenic in Older People; GRADE – Grading of Recommendations, Assessment, Development and Evaluations.

Ethics approval and consent to participate: Ethical approval will not be required as only published data will be used.

Consent for publication: Consent for publication will not be required as only published data will be used.

Availability of data and materials: The data are available upon request from the corresponding author.

Competing interests: The authors declare that they have no competing interests.

Funding: Not applicable.

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Authors' contributions: GVS and EAS conceived the study idea. GVS, EAS, and MN contributed to the design of the systematic review. GVS, EAS, and MN contributed to the data analysis plan. All authors contributed to the writing and editing of the manuscript and approved the final manuscript.

Acknowledgements: The authors are grateful to the Instituto Federal Goiano for support.

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with systematic review protocol submissions to BioMed Central journals from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

An Editorial from the Editors-in-Chief of *Systematic Reviews* details why this checklist was adapted - **Moher D, Stewart L & Shekelle P:** Implementing PRISMA-P: recommendations for prospective authors. *Systematic Reviews* 2016 **5**:15

Section/topic	#	Checklist item	Information reported		c
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	X		
Update	1b	If the protocol is for an update of a previous systematic review, identify as such		X	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract		X	
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	X		
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	X		
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments		X	
Support					
Sources	5a	Indicate sources of financial or other support for the review	X		
Sponsor	5b	Provide name for the review funder and/or sponsor	X		
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	X		
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	X		

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Section/topic	#	Checklist item	Information reported		c
			Yes	No	
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	X		
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	X		
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	X		
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	X		
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	X		
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	X		
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	X		
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	X		
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	X		
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	X		
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	X		
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	X		
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	X		
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	X		

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Section/topic	#	Checklist item	Information reported		c
			Yes	No	
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	X		
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	X		

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EFFECTIVENESS OF EXERCISE FOR OSTEOSARCOPENIA IN OLDER ADULTS: A SYSTEMATIC REVIEW PROTOCOL

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-045604.R1
Article Type:	Protocol
Date Submitted by the Author:	13-Apr-2021
Complete List of Authors:	Vinícius-Souza, Guilherme; Universidade Federal de Goias, Faculdade de Medicina Noll, Matias; Instituto Federal Goiano; University of Southern Denmark Silveira, Erika; Universidade Federal de Goias, Faculdade de Medicina
Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Evidence based practice, Geriatric medicine, Sports and exercise medicine
Keywords:	GERIATRIC MEDICINE, Musculoskeletal disorders < ORTHOPAEDIC & TRAUMA SURGERY, SPORTS MEDICINE

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EFFECTIVENESS OF EXERCISE FOR OSTEOSARCOPENIA IN OLDER ADULTS: A SYSTEMATIC REVIEW PROTOCOL

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EFFECTIVENESS OF EXERCISE FOR OSTEOSARCOPENIA IN OLDER ADULTS: A SYSTEMATIC REVIEW PROTOCOL

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ABSTRACT

Introduction: Osteosarcopenia is defined as the concomitant occurrence of sarcopenia and osteopenia or osteoporosis. Older adults with this syndrome have greater fragility and mortality risk compared to those without these conditions. Based on separate interventions with individuals with sarcopenia and osteoporosis, exercise has been recommended as a treatment for osteosarcopenia. However, there is no evidence of its efficacy. Our objective is to identify whether physical exercise can improve osteosarcopenia in older adults and lead to good health outcomes. **Methods and analysis:** We will perform a systematic review in the following databases: PubMed, Embase, Cochrane, and Scopus. The criterion of inclusion will be clinical trials involving physical exercise interventions in older adults diagnosed with osteosarcopenia. To assess the risk of bias, the Grading of Recommendations, Assessment, Development and Evaluation and Downs and Black tools will be used. For each search result, the quality of the evidence will ultimately receive one of four grades: high, moderate, low, or very low. The outcome of this study is to demonstrate the effectiveness of physical exercise in improving the parameters that lead to the diagnosis of osteosarcopenia (bone mineral density, quality of muscle mass, muscle strength, and physical function) in older adults. The possibility of meta-analysis will be assessed according to the homogeneity of the studies, using the methods of fixed or random effects. Sensitivity analyses will be performed, and the funnel plot will be used to assess publication bias. The proposed statistical analyses will be performed using STATA software, version 14.0. **Ethics and dissemination**: The results of the systematic review will be disseminated via publication in a peer-reviewed journal and presented at a relevant conference. As we will not use individual patient data, ethical approval is not required.

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Systematic review registration: CRD42020215659

Keywords: Older adults. Osteoporosis. Sarcopenia.

Strengths and limitations of this study

- Four important databases (PubMed [National Library of Medicine], Embase, Cochrane, and Scopus) will be analyzed.
- Our investigation will seek to determine the duration, frequency, intensity, and type of physical exercise that is most appropriate for preventing and treating osteosarcopenia.
- There may be high heterogeneity owing to the various diagnostic criteria and cutoff points for sarcopenia.
- There will be no language bias as our search will not place restrictions on language of publication.
- The results will contribute to the development of health prevention and promotion programs and may fill the gap pertaining to accurate physical exercise prescription in older adults with osteosarcopenia.

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Introduction

A recently defined syndrome [1], osteosarcopenia refers to the presence of sarcopenia plus osteopenia or osteoporosis in the same individual [2]. The diagnosis of osteopenia/osteoporosis requires an assessment of bone mineral density (BMD) through dual energy X-ray absorptiometry (DEXA) [3,4]. Sarcopenia, on the contrary, is characterized by low muscle strength, with diagnosis confirmed through the detection of low muscle quality and severity classified based on the weakness of physical performance [5].

The prevalence of osteosarcopenia in older adults varies by country. A study in Germany reported a prevalence of 28% [6], while in a study of Australian community dwellers, the rate was 37% [7]. In addition, when compared to older adults with isolated conditions, that is, sarcopenia, osteopenia, or osteoporosis, those with osteosarcopenia have more fractures [8] and a 15.1% higher risk of mortality [9].

Physical exercise (PE) has been recommended as a preventive strategy or non-pharmacological therapeutic approach for osteosarcopenia [10–12]. However, PE is prescribed in the context of isolated conditions, failing to consider osteosarcopenia as a unique condition that encompasses sarcopenia plus osteopenia/osteoporosis [11,13,14]. While there is a consensus in the literature on the effectiveness of PE for older adults with sarcopenia and osteoporosis [15–18], few studies have explored the efficacy of PE in people with osteosarcopenia.

Regarding osteosarcopenia in older adults, we were unable to identify any literature review addressing the effects of regular engagement in PE. Therefore, our aim is to develop a systematic review of the literature that can answer the following question: “What are the influences of regular PE practice on osteosarcopenia in older adults?” The

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results of this study may promote the application of PE as a form of non-pharmacological or complementary therapy for the prevention or treatment of osteosarcopenia.

Methods and analysis

This systematic review will be conducted according to the criteria of the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) [19,20]. We adopted the "PICO" structure and defined it as follows: "P" corresponds to older adults, "I" to physical exercise, "C" to the absence of physical exercise, and "O" to osteosarcopenia.

Eligibility criteria

Controlled, randomized, blind or open clinical trials, without language and publication period restrictions, that perform any type of minimum four-week PE intervention with older adults (aged ≥ 65 years) of both sexes diagnosed with concomitant osteopenia/osteoporosis and sarcopenia will be included.

We will consider individuals with osteopenia with a T-score < -1 in measurements performed by DEXA on the femoral neck, lumbar spine, or total hip, according to the World Health Organization's diagnostic criteria for BMD [3].

Sarcopenia has various definitions and diagnostic criteria [5,21]. Therefore, the criteria and cutoff points already proposed in studies that perform musculoskeletal measurements will be used [5,21–23], and studies with older adults with reduced muscle strength and mass will be included. Studies carried out with older adults with osteosarcopenic obesity will also be included in this review.

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However, studies with hospitalized older adults, with a focus on specific conditions, for example, stroke, will be excluded. We will also exclude animal studies, observational studies, opinion articles, editorials, narrative reviews, case series and comments, and duplicate studies. Publications that meet the inclusion criteria but for which the results are unavailable even after consulting the authors will be excluded.

Database search

Our search will be conducted in four databases (PubMed (National Library of Medicine), Embase, Cochrane, and Scopus). In addition, the GreyNet International platform and Google Scholar will be used to locate gray literature. To complement the search and ensure the saturation of the literature, the references of the selected articles will also be considered.

The search terms have been selected using Medical Subject Headings (MeSH) for the PubMed database. Search: ((Osteosarcopenia [Title/Abstract]) OR (Sarco-osteopenia [Title/Abstract]) OR (Osteo-sarcopenia [Title/Abstract]) OR (Osteosarcopenic [Title/Abstract]) AND (((((exercise [MeSH Terms]) OR (exercise [Title/Abstract])) OR (physical activity [Title/Abstract])) OR (physical activities [Title/Abstract])) OR ("Physical Fitness" [Title/Abstract]))) AND (((((((aged [MeSH Terms]) OR (aged [Title/Abstract])) OR (elderly [Title/Abstract])) OR (older adult [Title/Abstract])) OR (older adults [Title/Abstract])) OR (elder [Title/Abstract])) OR (Ageing [Title/Abstract])) OR (aging [Title/Abstract])). The same search model will be adapted to Embase, Cochrane, and Scopus databases (supplementary file). Searches will be carried out in April 2021.

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Review process

The search for articles will be carried out by two independent researchers (GVS and MN) and a third senior reviewer (EAS). The inclusion of articles will be based on a reading of the titles, and later, the abstracts. Finally, the complete content will be analyzed for inclusion.

After executing the search strategy, articles will be collated, and duplicates will be removed using Mendeley software. Then, two reviewers (GVS and MN) will independently screen the titles and abstracts of all articles identified in the literature search. Disagreements regarding inclusion will be discussed and resolved by a third reviewer (EAS). Both reviewers will perform the screening process using Rayyan software [24]. Inter-rater reliability for individual component ratings will be determined by calculating the percentage of agreement and the Cohen's Kappa coefficient [25]. The remaining articles will be read in full and evaluated to determine their eligibility based on the inclusion and exclusion criteria. Finally, eligible articles will be included in the systematic review. The entire process of selecting articles for the systematic review will be finalized by the end of September 2021.

According to PRISMA-P recommendations, we will prepare a flowchart with information about the screening of studies, reasons for inclusion and exclusion, and recording and viewing of this process.

Data extraction and study quality assessment

Data extraction and study quality assessment will be performed by two reviewers (GVS and MN); disagreements will be resolved by a third reviewer (EAS). To extract data from an article, a standardized form prepared by the authors will be used. The

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following aspects will be considered: author/year of publication, place of study, age group studied, sample size, intervention performed, time of intervention, and main results.

Risk of bias will be assessed using the 27-item Downs and Black instrument [26]. A score will be calculated for each manuscript (0–27 points) and scores above 70% will be used to define low risk of bias.

The Grading of Recommendations, Assessment, Development and Evaluation [27] system will be used to assess the quality of evidence. For each research result, the quality of the evidence will be classified as one of the following: high, moderate, low, or very low [28].

We will also analyze whether the authors of the included studies have addressed the impact of possible conflicts of interest and information regarding ethical approval [29]. We intend to finalize the data extraction process and evaluate the quality of the studies at the beginning of April 2021.

Data analysis

The outcome of this study is improvement in osteosarcopenia (BMD, appendicular muscle mass, muscle strength, and function) in older adults. For subgroup analysis, we will consider the following variables: sex, age group, and body mass index. We will group the results in meta-analysis of random effects and these will be reported using a forest plot to show the grouped effect of the findings. In addition, we will use a funnel plot for data accuracy assessment [30]. The percentage of variability attributable to heterogeneity will be estimated using the I2 test. I2 values of < 40, 40–60, 60–90, and > 90% correspond, respectively, to “not important,” “moderate,” “substantial,” and “considerable” levels of

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heterogeneity [30]. The proposed statistical analyses will be performed using STATA software, version 14.0.

Patient and Public Involvement

No patients involved.

Discussion

PE prescriptions are generally based on specific recommendations for osteoporosis and sarcopenia [10,11]. This is why our investigation will seek to determine the specific duration, frequency, intensity, and type of PE that is most appropriate for preventing and treating osteosarcopenia. We will seek to broaden the research by addressing the potential differences associated with varying exercise modalities: low or high-impact aerobics, resistance exercise, balance exercises, combined exercises, and whole-body vibration.

Through this protocol, it will be possible to conduct a systematic review and meta-analysis to elucidate the type of PE that is most effective for the prevention and/or treatment of osteosarcopenia in older adults. At present, we are not aware of a published systematic review on this topic. Consequently, the results may hold important implications for the field of gerontology. At the conclusion of this project, we aim to clarify the influence of PE on the parameters of osteosarcopenia, to trace paths of recommendations for possible therapeutic practices, and to identify the need for new studies.

Ethics and dissemination

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The results of the systematic review will be disseminated via publication in a peer-reviewed journal and presented at a relevant conference. As the data will not include individual patient data, ethical approval is not required.

Abbreviations: BMD – Bone mineral density; DEXA – Dual energy X-ray absorptiometry; PE – Physical exercise; PRISMA-P – Preferred Reporting Items for Systematic review and Meta-Analysis Protocols.

Ethics approval and consent to participate: Ethical approval will not be required as only published data will be used.

Consent for publication: Consent for publication will not be required as only published data will be used.

Availability of data and materials: The data are available upon request from the corresponding author.

Competing interests: The authors declare that they have no competing interests.

Funding: Not applicable.

Authors' contributions: GVS and EAS conceived the study idea. GVS, EAS, and MN contributed to the design of the systematic review. GVS, EAS, and MN contributed to the data analysis plan. All authors contributed to the writing and editing of the manuscript and approved the final manuscript.

Acknowledgments: The authors are grateful to the Instituto Federal Goiano for support.

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Table 1- Example of search strategy

1.	Osteosarcopenia OR Sarco-osteopenia OR Osteo-sarcopenia OR Osteosarcopenic
2.	exercise OR physical activity OR physical activities OR Physical Fitness
3.	Aged OR elderly OR older adult OR older adults OR elder OR Ageing OR Aging
4.	1 AND 2 AND 3

Table 2 - Example of search strategy for PubMed

	PUBMED	Results
1.	(((Osteosarcopenia[Title/Abstract]) OR (Sarco-osteopenia[Title/Abstract])) OR (Osteo-sarcopenia[Title/Abstract])) OR (Osteosarcopenic[Title/Abstract])	158
2.	((((exercise[MeSH Terms]) OR (exercise[Title/Abstract])) OR (physical activity[Title/Abstract])) OR (physical activities[Title/Abstract])) OR (Physical Fitness[Title/Abstract])	452,371
3.	Aged OR OR elderly OR older adult OR older adults OR elder OR Ageing OR Aging	3,812,703
4.	1 AND 2 AND 3	41
5.	1 AND 2 AND 3: Filters applied: Clinical Trial, Humans	3

Table 3 - Example of search strategy Embase

	EMBASE	Results
1.	osteosarcopenia:ab,ti OR 'sarco osteopenia':ab,ti OR 'osteo sarcopenia':ab,ti OR osteosarcopenic:ab,ti	253
2.	exercise:ab,ti OR 'physical activity':ab,ti OR 'physical activities':ab,ti OR 'physical fitness':ab,ti	495,253
3.	elderly:ab,ti OR 'older adult':ab,ti OR 'older adults':ab,ti OR aged:ab,ti OR ageing:ab,ti OR aging:ab,ti	1,379,931
4.	#1 And #2 And #3	40
5.	#1 And #2 And #3 - Filters apllied: Humans	34

Table 4 - Example of search strategy Scopus

	SCOPUS	Results
1.	Osteosarcopenia OR Sarco-osteopenia OR Osteo-sarcopenia OR Osteosarcopenic AND exercise OR physical activity OR physical activities OR physical fitness AND aged OR elderly OR older adult OR older adults OR elder OR ageing OR aging	113
2.	Filters apllied: article	71

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Table 5 - Example of search strategy Cochrane Library

	COCHRANE	Results
1.	(osteosarcopenia):ti,ab,kw OR (sarco-osteopenia):ti,ab,kw OR (osteo-sarcopenia):ti,ab,kw OR (osteosarcopenic):ti,ab,kw	
2.	(exercise):ti,ab,kw OR (physical activity):ti,ab,kw OR (physical activities):ti,ab,kw OR (physical fitness):ti,ab,kw	
3.	(aged):ti,ab,kw OR (elderly):ti,ab,kw OR (older adult):ti,ab,kw OR (older adults):ti,ab,kw OR (elder):ti,ab,kw	
4.	#1 And #2 And #3	14

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